

The vector and Banach lattices library

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April 22, 2026

Chapter 1

The Basics

The results the reader may find missing in this chapter are likely already part of `mathlib`. Some statements may have an “unusually formal” formulation. The reason is that we try to keep the blueprint as similar as possible to the code. The blueprint is a compromise between the Lean code and the informal human maths.

1.1 Vector lattices and the Archimedean property

This section sets up the basic order-theoretic algebra underpinning the rest of the library. We work first in the generality of a lattice-ordered abelian group, establishing the fundamental identities for the positive part x^+ , the negative part x^- , and the absolute value $|x|$, together with the uniqueness of the positive part and the interaction of the lattice operations with translation and with arbitrary suprema and infima. We then specialise to *vector lattices*, i.e. real vector spaces with a compatible lattice structure, and record how non-negative scalars distribute over the lattice operations, culminating in the identity $|\lambda x| = |\lambda| |x|$. The section closes with the Archimedean property, which plays a foundational role in later chapters: we give two equivalent formulations and use them to show that an Archimedean vector lattice of rank at least two necessarily has vectors which are neither positive nor negative.

1.1.1 Lattice-ordered abelian groups

Throughout this subsection X denotes an additive abelian group equipped with a lattice order compatible with addition (an ℓ -group). We write $x^+ := x \sqcup 0$, $x^- := (-x) \sqcup 0$ and $|x| := x \sqcup (-x)$.

Lemma 1.1.1. *Let X be a lattice-ordered abelian group and let $x \in X$. Then $|x| = 0$ if and only if $x = 0$.*

Lemma 1.1.2. *Let X be a lattice-ordered abelian group, let $x \in X$, and suppose $u, v \in X$ satisfy $x = u - v$ and $u \sqcap v = 0$. Then $u = x^+$.*

Lemma 1.1.3. *For all x, y in a lattice-ordered abelian group, $x \sqcup y = x + (y - x)^+$.*

Lemma 1.1.4. *For all x, y in a lattice-ordered abelian group, $x \sqcap y = x - (x - y)^+$.*

Lemma 1.1.5. *For all x, y in a lattice-ordered abelian group, $x - x \sqcap y = (x - y)^+$.*

Lemma 1.1.6. *For all x, y in a lattice-ordered abelian group, $(x + y)^+ \leq x^+ + y^+$.*

Lemma 1.1.7. Let X be a lattice-ordered abelian group and let $x, a, b \in X$ with $0 \leq x$, $0 \leq a$, and $0 \leq b$. Then $x \sqcap (a + b) \leq x \sqcap a + x \sqcap b$.

Lemma 1.1.8. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x + a = \sup\{x + z : z \in A\}$.

Lemma 1.1.9. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x + a = \inf\{x + z : z \in A\}$.

Lemma 1.1.10. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x \sqcap a = \sup\{x \sqcap z : z \in A\}$.

Lemma 1.1.11. Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x \sqcup a = \inf\{x \sqcup z : z \in A\}$.

Lemma 1.1.12. Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a + b = \inf\{p + q : p \in A, q \in B\}$.

Lemma 1.1.13. Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \sqcup b = \inf\{p \sqcup q : p \in A, q \in B\}$.

Lemma 1.1.14. Let X be a lattice, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \sqcap b = \inf(A \cup B)$.

1.1.2 Vector lattices

A *vector lattice* is a lattice-ordered abelian group equipped with a real vector space structure whose scalar multiplication by non-negative reals is monotone. Throughout this subsection X denotes a vector lattice.

Definition 1.1.15. Let X be an additive abelian group equipped with a lattice structure compatible with addition. A *vector lattice* structure on X consists of a real vector space structure on X for which scalar multiplication by non-negative reals is monotone: for all $\lambda \geq 0$ and $x \leq y$ in X , $\lambda x \leq \lambda y$.

Lemma 1.1.16. Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \sqcup y) = (\lambda x) \sqcup (\lambda y)$.

Lemma 1.1.17. Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \sup A$, then $\lambda a = \sup\{\lambda z : z \in A\}$.

Lemma 1.1.18. Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \inf A$, then $\lambda a = \inf\{\lambda z : z \in A\}$.

Lemma 1.1.19. Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \sqcap y) = (\lambda x) \sqcap (\lambda y)$.

Lemma 1.1.20. Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \sqcup \beta) \cdot x = (\alpha x) \sqcup (\beta x)$.

Lemma 1.1.21. Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \sqcap \beta) \cdot x = (\alpha x) \sqcap (\beta x)$.

Theorem 1.1.22. Let X be a vector lattice, let $x \in X$ and $\alpha \in \mathbb{R}$. Then $|\alpha x| = |\alpha| \cdot |x|$.

Lemma 1.1.23. Let X be a vector lattice, let $x, y \in X$ with $x \sqcap y = 0$ and let $\lambda \in \mathbb{R}$ with $0 \leq \lambda$. Then $(\lambda x) \sqcap y = 0$.

1.1.3 The Archimedean property

Definition 1.1.24. A lattice-ordered abelian group X is called *Archimedean* when, for every $x, y \in X$ with $0 \leq x$ and $n \cdot x \leq y$ for all $n \in \mathbb{N}$, one has $x = 0$.

Lemma 1.1.25. Let X be a lattice-ordered abelian group. Then X is Archimedean if and only if for all $x, y \in X$ the condition $n \cdot x \leq y$ for all $n \in \mathbb{N}$ implies $x \leq 0$.

Lemma 1.1.26. A vector lattice X is Archimedean if and only if for every $u \in X$ with $0 \leq u$, $\inf\{n^{-1} \cdot u : n \in \mathbb{N}, n > 0\} = 0$.

Lemma 1.1.27. Let X be an Archimedean lattice-ordered abelian group and let $x, y \in X$. If $n \cdot |x| \leq y$ for all $n \in \mathbb{N}$, then $x = 0$.

Theorem 1.1.28. Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exists $x \in X$ which is neither non-negative nor non-positive.

1.2 Riesz decomposition

This section collects the Riesz decomposition theorem and its standard refinements in a lattice-ordered abelian group X . The binary form splits a positive element dominated by a sum of two positive elements into a corresponding sum. Iterating and combining the argument yields a matrix form for two partitions of the same positive sum, a version with a finite sum on the left hand side, a decomposition adapted to the absolute value, and an identification of sum intervals $[0, x] + [0, y] = [0, x + y]$ for $x, y \geq 0$.

Theorem 1.2.1. Let X be a lattice-ordered abelian group and let $x, y, z \in X$ with $0 \leq x$, $0 \leq y$, $0 \leq z$ and $x \leq y + z$. Then there exist $x_1, x_2 \in X$ with $0 \leq x_1 \leq y$, $0 \leq x_2 \leq z$ and $x = x_1 + x_2$.

Theorem 1.2.2. Let X be a lattice-ordered abelian group, let $m, n \in \mathbb{N}$, let $u : \{0, \dots, m-1\} \rightarrow X$ and $z : \{0, \dots, n-1\} \rightarrow X$ with $0 \leq u_i$ for all i and $0 \leq z_j$ for all j , and $\sum_i u_i = \sum_j z_j$. Then there exists a family $x : \{0, \dots, m-1\} \times \{0, \dots, n-1\} \rightarrow X$ with $0 \leq x_{ij}$ for all i, j such that $u_i = \sum_j x_{ij}$ for every i and $z_j = \sum_i x_{ij}$ for every j .

Theorem 1.2.3. Let X be a lattice-ordered abelian group, let $k \in \mathbb{N}$, let $u : \{0, \dots, k-1\} \rightarrow X$ and $x, y \in X$ with $0 \leq u_i$ for all i , $0 \leq x$, $0 \leq y$ and $\sum_i u_i \leq x + y$. Then there exist $v, w : \{0, \dots, k-1\} \rightarrow X$ with $0 \leq v_i$ and $0 \leq w_i$ for all i , $\sum_i v_i \leq x$, $\sum_i w_i \leq y$, and $u_i = v_i + w_i$ for every i .

Theorem 1.2.4. Let X be a lattice-ordered abelian group, let $x, u, v \in X$ with $0 \leq u$, $0 \leq v$ and $|x| = u + v$. Then there exist $y, z \in X$ with $x = y + z$, $|y| = u$ and $|z| = v$.

Theorem 1.2.5. Let X be a lattice-ordered abelian group, let $n \in \mathbb{N}$, let $x \in X$ and let $v : \{0, \dots, n-1\} \rightarrow X$ with $0 \leq v_i$ for every i and $|x| \leq \sum_i v_i$. Then there exists $y : \{0, \dots, n-1\} \rightarrow X$ with $x = \sum_i y_i$ and $|y_i| \leq |x| \sqcap v_i$ for every i .

Theorem 1.2.6. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$ and $0 \leq y$. Then $[0, x] + [0, y] = [0, x + y]$ as subsets of X .

1.3 Normed and Banach lattices

A *normed vector lattice* is a real vector lattice equipped with a norm satisfying the solid axiom $|x| \leq |y| \Rightarrow \|x\| \leq \|y\|$. This compatibility between the norm and the lattice operations has a number of immediate consequences: the lattice operations $\sqcup$, $\sqcap$ and $|\cdot|$ are norm-continuous, the space is automatically Archimedean, the positive cone is norm-closed, order intervals are norm-closed and norm-bounded, and monotone norm-convergent sequences converge to their supremum. A *Banach lattice* is a normed vector lattice whose norm is complete.

Definition 1.3.1. Let X be a normed abelian group equipped with a compatible lattice structure. A *normed vector lattice* structure on X consists of a vector lattice structure together with the solid axiom $|x| \leq |y| \Rightarrow \|x\| \leq \|y\|$ for all $x, y \in X$, and the identity $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{R}$ and $x \in X$.

Lemma 1.3.2. Every normed vector lattice is a normed space over $\mathbb{R}$.

Theorem 1.3.3. Let X be a normed vector lattice. The map $(x, y) \mapsto x \sqcup y$ from $X \times X$ to X is continuous.

Theorem 1.3.4. Let X be a normed vector lattice. The map $(x, y) \mapsto x \sqcap y$ from $X \times X$ to X is continuous.

Theorem 1.3.5. Let X be a normed vector lattice. The map $x \mapsto |x|$ from X to X is Lipschitz with constant 1.

Theorem 1.3.6. Every normed vector lattice is Archimedean.

Lemma 1.3.7. Let X be a normed vector lattice. The order relation on X is closed in the norm topology.

Theorem 1.3.8. Let X be a normed vector lattice. The positive cone $\{x \in X : 0 \leq x\}$ is norm-closed.

Theorem 1.3.9. Let X be a normed vector lattice, let $u, v : \mathbb{N} \rightarrow X$ and $a, b \in X$. If $u_n \rightarrow a$ and $v_n \rightarrow b$ in norm and $u_n \leq v_n$ for every n , then $a \leq b$.

Theorem 1.3.10. Let X be a normed vector lattice, let $u : \mathbb{N} \rightarrow X$ be monotone and let $l \in X$. If $u_n \rightarrow l$ in norm, then l is the least upper bound of the range of u .

Theorem 1.3.11. Let X be a normed vector lattice, let $u : \mathbb{N} \rightarrow X$ be antitone and let $l \in X$. If $u_n \rightarrow l$ in norm, then l is the greatest lower bound of the range of u .

Lemma 1.3.12. Let X be a normed vector lattice and let $a, b \in X$. The order interval $[a, b]$ is norm-closed.

Lemma 1.3.13. Let X be a normed vector lattice, let $a, b \in X$ and let $x \in [a, b]$. Then $\|x\| \leq \|a\| \sqcup \|b\|$.

Theorem 1.3.14. Let X be a normed vector lattice and let $s \subseteq X$ be both bounded below and bounded above in the order. Then s is norm-bounded.

Definition 1.3.15. A *Banach lattice* is a normed vector lattice whose underlying normed space is complete.

1.4 Disjointness

Two elements of a lattice-ordered abelian group are *disjoint* when the infimum of their absolute values vanishes. This section develops the elementary calculus of disjointness: symmetry, the zero rules, the Birkhoff identity $|x + y| = |x| + |y|$ for disjoint pairs, uniqueness of the positive/negative decomposition, compatibility with scalar multiplication and with monotonicity under the absolute value, and closure under finite suprema and sums. From these one deduces the finite-family identity $|\sum_i \alpha_i x_i| = \sum_i |\alpha_i| |x_i|$ for pairwise-disjoint families, from which any pairwise-disjoint family of non-zero vectors is linearly independent over $\mathbb{R}$. The section closes with a normed-lattice statement: a norm limit of a pairwise-disjoint sequence must be zero.

Definition 1.4.1. Let X be a lattice-ordered abelian group and let $x, y \in X$. We say x and y are *disjoint* when $|x| \sqcap |y| = 0$.

Lemma 1.4.2. Let X be a lattice-ordered abelian group and let $x, y \in X$. Then x and y are disjoint if and only if y and x are disjoint.

Lemma 1.4.3. Let X be a lattice-ordered abelian group and let $x \in X$. Then 0 and x are disjoint.

Lemma 1.4.4. Let X be a lattice-ordered abelian group and let $x \in X$. Then x and 0 are disjoint.

Lemma 1.4.5. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$ and $0 \leq y$. If x and y are disjoint, then $x \sqcap y = 0$.

Lemma 1.4.6. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $x \sqcap y = 0$. Then x and y are disjoint.

Theorem 1.4.7. Let X be a lattice-ordered abelian group and let $u_1, v_1, u_2, v_2 \in X$ be non-negative elements. If u_1 and v_1 are disjoint, u_2 and v_2 are disjoint, and $u_1 - v_1 = u_2 - v_2$, then $u_1 = u_2$ and $v_1 = v_2$.

Theorem 1.4.8. Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x + y| = |x| + |y|$.

Lemma 1.4.9. Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$, $0 \leq y$ and x, y disjoint. Then $|x \sqcup y| = |x| \sqcup |y|$.

Lemma 1.4.10. Let X be a lattice-ordered abelian group and let $x \in X$. Then x^+ and x^- are disjoint.

Lemma 1.4.11. Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and $|z| \leq |y|$, then x and z are disjoint.

Lemma 1.4.12. Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and $|z| \leq |x|$, then z and y are disjoint.

Lemma 1.4.13. Let X be a vector lattice and let $x, y \in X$ be disjoint. For every $\alpha \in \mathbb{R}$, αx and y are disjoint.

Lemma 1.4.14. Let X be a vector lattice and let $x, y \in X$ be disjoint. For every $\alpha \in \mathbb{R}$, x and αy are disjoint.

Theorem 1.4.15. Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exist non-zero $u, v \in X$ with u and v disjoint.

Lemma 1.4.16. *Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$, $0 \leq y$ and x, y disjoint. Then $x + y = x \sqcup y$.*

Lemma 1.4.17. *Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x| \sqcup |y| = |x| + |y|$.*

Lemma 1.4.18. *Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x + y| = |x| \sqcup |y|$.*

Lemma 1.4.19. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and x and z are disjoint, then x and $y \sqcup z$ are disjoint.*

Lemma 1.4.20. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and x and z are disjoint, then x and $y + z$ are disjoint.*

Lemma 1.4.21. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and z are disjoint and y and z are disjoint, then $x \sqcup y$ and z are disjoint.*

Lemma 1.4.22. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and z are disjoint and y and z are disjoint, then $x + y$ and z are disjoint.*

Lemma 1.4.23. *Let X be a lattice-ordered abelian group and let $x, y \in X$. Then $x - x \sqcap y$ and $y - x \sqcap y$ are disjoint.*

Theorem 1.4.24. *Let X be a vector lattice, let $x, y \in X$ and $\lambda \in \mathbb{R}$ with $0 < \lambda$. Then $(x - \lambda y)^+$ and $(y - \lambda^{-1}x)^+$ are disjoint.*

Theorem 1.4.25. *Let X be a lattice-ordered abelian group, let $n \in \mathbb{N}$ with $n > 0$, and let $f : \{0, \dots, n-1\} \rightarrow X$ be a family of non-negative elements which is pairwise disjoint. Then $\sum_i f_i = \bigsqcup_i f_i$.*

Theorem 1.4.26. *Let X be a vector lattice, let $n \in \mathbb{N}$, and let $x : \{0, \dots, n-1\} \rightarrow X$ be a pairwise-disjoint family. For every $\alpha : \{0, \dots, n-1\} \rightarrow \mathbb{R}$, $|\sum_i \alpha_i x_i| = \sum_i |\alpha_i| |x_i|$.*

Theorem 1.4.27. *Let X be a vector lattice, let $n \in \mathbb{N}$ with $n > 0$, and let $x : \{0, \dots, n-1\} \rightarrow X$ be a pairwise-disjoint family of non-negative elements. For every non-negative family $\alpha : \{0, \dots, n-1\} \rightarrow \mathbb{R}$, $\sum_i \alpha_i x_i = \bigsqcup_i \alpha_i x_i$.*

Theorem 1.4.28. *Let X be a vector lattice, let $n \in \mathbb{N}$, and let $x : \{0, \dots, n-1\} \rightarrow X$ be a pairwise-disjoint family of non-zero elements. Then x is linearly independent over $\mathbb{R}$.*

Theorem 1.4.29. *Let Y be a normed vector lattice, let $u : \mathbb{N} \rightarrow Y$ be a pairwise-disjoint sequence, and let $x \in Y$. If $u_n \rightarrow x$ in norm, then $x = 0$.*

Chapter 2

Substructures

2.1 Vector sublattices

A *vector sublattice* of a vector lattice X is a linear subspace closed under the lattice operations. Since the lattice operations are inter-definable, closure under any one of them is enough: closure under $\sqcup$ or under $|\cdot|$ both characterise the notion. This section introduces the bundled structure, shows that every vector sublattice is a vector lattice in its own right and that the collection of vector sublattices of a fixed ambient space forms a complete lattice under inclusion, and describes the vector sublattice generated by a set in several concrete situations — a submodule, a pointed cone, and a pairwise-disjoint family of non-negative vectors. The section closes with the normed setting: the induced normed vector lattice structure on the subtype of a vector sublattice, its topological closure, completeness when the ambient space is a Banach lattice, and separability of the closed vector sublattice generated by a countable subset.

2.1.1 Pointed cones

Before introducing vector sublattices themselves, we record that in a vector lattice the sup-closure and inf-closure of a pointed cone are again pointed cones. These two operations will appear as basic building blocks in the descriptions of the generated vector sublattice. Recall that a *pointed cone* is a subset of a vector subspace that is closed under addition, positive scalar multiplication, and contains the origin.

Lemma 2.1.1. *Let X be a vector lattice and let C be a pointed cone in X . Then the sup-closure of C , viewed as a subset of X , is a pointed cone.*

Lemma 2.1.2. *Let X be a vector lattice and let C be a pointed cone in X . Then the inf-closure of C , viewed as a subset of X , is a pointed cone.*

2.1.2 Vector sublattices

Throughout the rest of the section X denotes a vector lattice.

Definition 2.1.3. Let X be a vector lattice. A *vector sublattice* of X is a linear subspace $Y \subseteq X$ which is closed under $\sqcup$: for all $x, y \in Y$, $x \sqcup y \in Y$.

Lemma 2.1.4. *Let Y be a vector sublattice of X and let $x, y \in Y$. Then $x \sqcap y \in Y$.*

Lemma 2.1.5. *Let Y be a vector sublattice of X and let $x \in Y$. Then $x^+ \in Y$.*

Lemma 2.1.6. *Let Y be a vector sublattice of X and let $x \in Y$. Then $x^- \in Y$.*

Lemma 2.1.7. *Let Y be a vector sublattice of X and let $x \in Y$. Then $|x| \in Y$.*

Lemma 2.1.8. *Let Y be a vector sublattice of X . Then the underlying set of Y is a sublattice of X .*

Since the lattice operations of a vector lattice are inter-definable with the absolute value, a linear subspace closed under $|\cdot|$ is automatically a vector sublattice.

Lemma 2.1.9. *Let M be a linear subspace of X which is closed under $|\cdot|$, i.e. $|x| \in M$ for every $x \in M$. Then M is the underlying submodule of a vector sublattice of X .*

Lemma 2.1.10. *Let M be a linear subspace of X . Then M is closed under $|\cdot|$ if and only if M is closed under $\sqcup$.*

A vector sublattice inherits a vector lattice structure from the ambient space, with the lattice operations, addition and scalar multiplication computed pointwise.

Lemma 2.1.11. *Let Y be a vector sublattice of X . The underlying subset of Y carries a vector lattice structure whose lattice operations, addition and scalar multiplication agree pointwise with those of X .*

The collection of vector sublattices of X is ordered by inclusion. It has a greatest element $\top = X$, a least element $\perp = \{0\}$, and is closed under arbitrary intersections.

Lemma 2.1.12. *The forgetful map from vector sublattices of X to linear subspaces of X is injective.*

Lemma 2.1.13. *The whole space X is the greatest vector sublattice of X , the zero subspace $\{0\}$ is the least vector sublattice of X , and the intersection of an arbitrary family of vector sublattices is again a vector sublattice of X .*

2.1.3 The vector sublattice generated by a set

For any subset $s \subseteq X$, there is a smallest vector sublattice of X containing s , obtained as the infimum of all vector sublattices that contain s . The subsection characterises this generated sublattice in several concrete situations.

Definition 2.1.14. Let s be a subset of X . The vector sublattice *generated* by s is the intersection of all vector sublattices of X containing s .

Lemma 2.1.15. *For every subset $s \subseteq X$, s is contained in the vector sublattice generated by s .*

Lemma 2.1.16. *Let $s \subseteq X$ and let Y be a vector sublattice of X with $s \subseteq Y$. Then the vector sublattice generated by s is contained in Y .*

Theorem 2.1.17. *Let M be a linear subspace of X . Then the underlying set of the vector sublattice generated by M equals the sup-closure of the inf-closure of M :*

$$\text{generated}(M) = \text{supClosure}(\text{infClosure}(M)).$$

Theorem 2.1.18. *Let M be a linear subspace of X . Then the underlying set of the vector sublattice generated by M equals the inf-closure of the sup-closure of M :*

$$\text{generated}(M) = \text{infClosure}(\text{supClosure}(M)).$$

Theorem 2.1.19. *Let C be a pointed cone in X . Then the underlying set of the vector sublattice generated by C equals the difference set of the sup-closure of C with itself:*

$$\text{generated}(C) = \text{supClosure}(C) - \text{supClosure}(C).$$

Theorem 2.1.20. *Let C be a pointed cone in X . Then the underlying set of the vector sublattice generated by C equals the difference set of the inf-closure of C with itself:*

$$\text{generated}(C) = \text{infClosure}(C) - \text{infClosure}(C).$$

Theorem 2.1.21. *Let $A \subseteq X$ be a set of pairwise-disjoint elements with $0 \leq a$ for every $a \in A$. Then the underlying submodule of the vector sublattice generated by A coincides with the linear span of A over $\mathbb{R}$.*

2.1.4 Normed and Banach sublattices

In the normed setting, a vector sublattice Y of a normed vector lattice X inherits a normed vector lattice structure on its subtype. The topological closure of a vector sublattice is again a vector sublattice; in a Banach lattice, norm-closed vector sublattices are themselves Banach lattices. Finally, the closed vector sublattice generated by a countable subset is separable.

Lemma 2.1.22. *Let X be a normed vector lattice and let Y be a vector sublattice of X . Then the underlying subset of Y is a normed vector lattice under the induced norm and lattice operations.*

Lemma 2.1.23. *Let X be a normed vector lattice and let Y be a vector sublattice of X . Then the norm closure of Y is a vector sublattice of X .*

Lemma 2.1.24. *Let X be a Banach lattice and let Y be a norm-closed vector sublattice of X . Then Y is a Banach lattice under the induced structures.*

Theorem 2.1.25. *Let X be a normed vector lattice and let $A \subseteq X$ be countable. Then the topological closure of the vector sublattice generated by A is separable.*